

## Data Preparation

Created 3 grouped categorical variables by merging small/sparse levels to respect the **max 4 levels rule**:

- `purpose_3` (Communications / Earth Observation / Other)
- `country_4` (US / China / Europe / Rest of World)
- `users_4` (Commercial / Government-Civil / Military / Mixed)

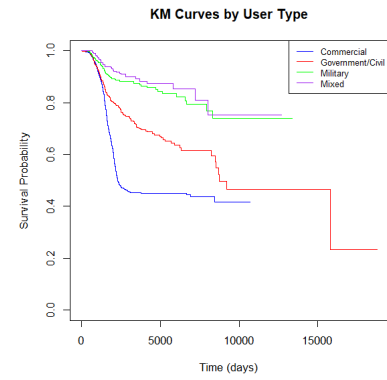
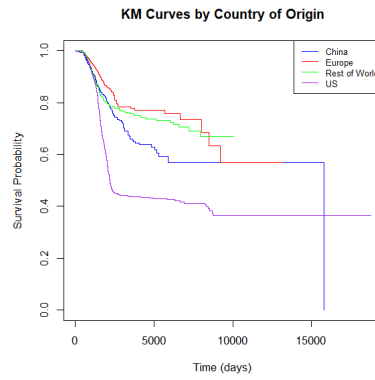
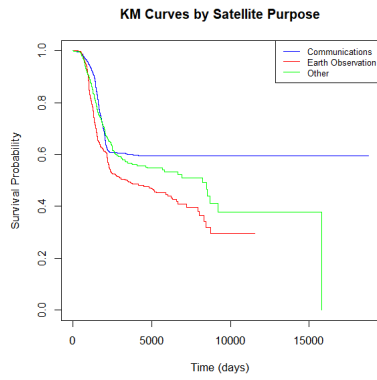
### For Each Categorical Variable:

1. **Plotted KM curves** — visualising survival differences across groups
2. **Extracted quartiles (Q1, Median, Q3)** with 95% confidence intervals
3. **Ran the Peto-Wilcoxon test** (justified by early differences and curve crossings)

### Key Findings

- **Purpose:** Communications satellites survive longest; Earth Observation decays faster
- **Country:** European and Rest of World satellites are most durable; US decays fastest (Starlink effect)
- **User type:** Military and Mixed survive best; Commercial decays fastest (again Starlink)
- **All three tests:**  $p < 2e-16$  — survival differences are overwhelmingly significant across all groupings

As I understand the professor asks to focus only on one categorical variable “in case you have several categorical variables, take the one that is most relevant to the data” so I did few that I thought are most relevant and we can decide as a team which one we keep. The results are below I will also send the r script if someone could push in git hub it will be easier for the rest



**Quartile Estimates (days) with 95% CIs**

| Group             | Q1   | Q1_lower | Q1_upper | Median | Med_lower | Med_upper | Q3    | Q3_lower | Q3_upper |
|-------------------|------|----------|----------|--------|-----------|-----------|-------|----------|----------|
| Communications    | 1744 | 1689     | 1806     | NA     | NA        | NA        | NA    | NA       | NA       |
| Earth Observation | 1309 | 1259     | 1366     | 3438   | 2451      | 5109      | NA    | 8446     | NA       |
| Other             | 1697 | 1513     | 1879     | 8249   | 5677      | NA        | 15781 | NA       | NA       |

| Group         | Q1   | Q1_lower | Q1_upper | Median | Med_lower | Med_upper | Q3    | Q3_lower | Q3_upper |
|---------------|------|----------|----------|--------|-----------|-----------|-------|----------|----------|
| China         | 2465 | 2180     | 3046     | 15781  | 5900      | NA        | 15781 | NA       | NA       |
| Europe        | 6649 | 2690     | NA       | NA     | 9205      | NA        | NA    | NA       | NA       |
| Rest of World | 3749 | 2500     | 7226     | NA     | NA        | NA        | NA    | NA       | NA       |
| US            | 1544 | 1520     | 1572     | 2183   | 2139      | 2263      | NA    | NA       | NA       |

| Group            | Q1   | Q1_lower | Q1_upper | Median | Med_lower | Med_upper | Q3    | Q3_lower | Q3_upper |
|------------------|------|----------|----------|--------|-----------|-----------|-------|----------|----------|
| Commercial       | 1580 | 1555     | 1605     | 2288   | 2217      | 2474      | NA    | NA       | NA       |
| Government/Civil | 2737 | 2370     | 3467     | 8727   | 8479      | NA        | 15781 | 15781    | NA       |
| Military         | 8317 | 6649     | NA       | NA     | NA        | NA        | NA    | NA       | NA       |
| Mixed            | NA   | 7226     | NA       | NA     | NA        | NA        | NA    | NA       | NA       |